

Overview

A measured path from charge to coupled fields

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

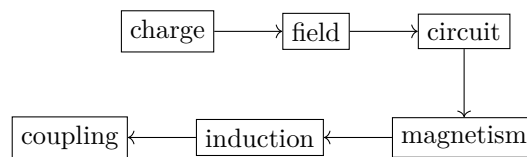
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- This overview
- A bound lab notebook
- The safety guide

Predict first

Which is more convincing: a named effect, a visible effect, or a measurement that changes predictably? Explain.



Investigate

1. Read Safety with the learner and assign roles.
2. Take Lessons 1–7 in order; they establish charge, field, measurement, induction, ratio, and stored energy.
3. Lessons 8–10 connect resonance, coupling, and historical discharge apparatus without crossing into learner high voltage.
4. Use demonstrations after their named prerequisite. Build the low-voltage coupled-coil instrument only after Lessons 1–9.

Explain from the evidence

The course treats a model as useful only when it predicts an observation. Each lesson follows the same loop: predict, assemble deenergized, adult inspection, energize briefly, measure, disconnect, explain. The final machine is not the purpose; it is a test of whether the learner can follow energy and fields across several representations.

Change one thing

Make a concept map after every lesson. Draw a solid arrow for a measured connection and a dotted arrow for a question. Revisit and revise rather than erase.

Evidence record

Prediction	_____
One change	_____
Observation	_____
Claim + because	_____

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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